



Unraveling the impact of travel time, cost, and transit burdens on commute mode choice for different income and age groups

Jaehyun Ha^a, Sugie Lee^{a,*}, Joonho Ko^b

^a Department of Urban Planning & Engineering, Hanyang University, Republic of Korea

^b Graduate School of Urban Studies, Hanyang University, Republic of Korea

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ABSTRACT

It is well known that faster and less-expensive transit systems with shorter walking distances, as well as less transfers, are associated with transit use. In this regard, recent studies have focused on estimating transit's competitiveness and further suggested possible policy interventions. However, there were not much attention on measuring travel time and cost differences between transit and auto by considering the temporal variability. In addition, only a limited study have focused on the impact of either walking during transfers on mode choice or the stage of walking that most affects mode choice. This paper provides additional insights into the impact of travel time, cost, and transit burdens on commute mode choice based on the analysis in a transit-oriented mega city, Seoul, Korea. Together with the 2016 household travel survey data, we utilize the web-based route guide information to measure time, cost, and transit burdens for both chosen and non-chosen modes. The results reveal that commuters are significantly affected by the travel time and cost difference between modes. Additionally, the results indicate that the odds of driving increases by 4.7% and 24.3% when commuters have to transfer once or more than once, respectively. Also, commuters were mostly disinclined to walk longer distances during transfer and egress. The effect of time, cost, and transit burdens on mode choice also varied by income and age groups, further implying the group of people that should be focused when devising policies to encourage transit use. To conclude, this paper suggests potential levers for transportation policies to trigger modal shifts toward sustainable modes by using advanced data sources that better estimate the travel characteristics by departure time based on a point-to-point travel survey data.

1. Introduction

In the past decades, numerous studies have examined the determinants of mode choice and further suggested potential policies that can help travelers alter their mode choice (Asensio, 2002; Chakrabarti, 2017). Most studies mainly addressed the impact of personal, household attributes, as well as the built environment, transit service, and self-preferences on mode choice (Cervero, 2002; Ding et al., 2017; Manaugh and El-Geneidy, 2015). The primary assumption of these research is that the travelers make their decision rationally, which is in line with the consumer choice theory (De Donnea, 1972). Meanwhile, although few studies argued that it is challenging to change commuters' mode choice regarding that those decisions are highly driven by habit (Schneider, 2013), most

* Corresponding author at: Science & Technology Bld. 303-1, Hanyang University, 222 Wangsimni-ro, Seongdong-gu, Seoul 04763, Republic of Korea.

E-mail addresses: jaehyunha@hanyang.ac.kr (J. Ha), sugielee@hanyang.ac.kr (S. Lee), jko@hanyang.ac.kr (J. Ko).

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studies and policy interventions have mainly focused on commuting travels (Wu and Hong, 2017; Ko et al., 2019). Within this context, there is no doubt that we should focus on auto-to-transit mode shift of commutes, which is one of people's daily routine. More in detail, several studies have reported that encouraging transit modes would mitigate problems such as air pollution, congestion, and traffic accidents and further bring health benefits (Chen and Jou, 2019; Zhang, 2010; Lachapelle and Frank, 2009).

When it comes to Seoul Metropolitan area (SMA), the city government has recently provided fare discount for transit users during early morning hours and improved transit systems by adding transfer centers and more exclusive median bus lanes on major arterials. Besides, the government has also offered real-time information about transit schedule and routes via various channels to travelers. These measures were designed to reduce travel time, cost, and transit burdens (e.g. access and egress distance, number of transfers) that are known to influence travelers' mode choice (Chakrabarti, 2017; Javanmardi et al., 2015). As a result, the modal share of transit in SMA has increased from 45.2% to 52.5% over the period between 2006 and 2016 based on the household travel survey (HTS) data, further indicating that such policy measures have been quite effective. Regarding the fact that the government is currently making the last-ditch effort to encourage transit use by constructing an express train that can operate at a speed of 100 km per hour, it is timely to examine how the transit's competitiveness relative to driving affects mode choice in Seoul, Korea. Here, transit's competitiveness is a widely used concept which implies that travelers select their mode by comparing the cost between two or more possible options (Schaller, 1999; Lee et al., 2019).

A number of studies have identified that faster and less-expensive transit systems with better accessibility are highly associated with transit use (Cervero, 2002; Frank et al., 2008; Chapman et al., 2007). Meanwhile, some recent studies have made efforts to better estimate transit competitiveness by using advanced data sources such as web-based route guide information (Chakrabarti, 2017; Golshani et al., 2018; Hasnine et al., 2019). It is worthwhile to note that transit-oriented mega cities have not been often studied. To elaborate, we expect that such an investigation on Seoul, Korea would help policymakers to formulate effective interventions that can maximize the level of transit use regarding travel time and cost. Additionally, there were not much attention on either walking during transfer or the stage of walking (e.g. access, egress, transfer) that most affects mode choice decisions. Another significant knowledge gap is about the group of people which are most sensitive to the factors such as time, cost, and transit burden when selecting their modes. More in detail, a few studies have shown that travelers of different age and income levels have diverse preferences, lifestyles, and attitudes, which could lead to different travel behavior patterns (Cheng et al., 2017; Chang, 2013; Figueroa et al., 2014; Haustein, 2012). In this regard, it is necessary to examine the heterogeneity among different groups by socioeconomic status, mainly focusing on their response to travel time, cost, and transit burdens when selecting travel mode between driving and public transit. Keeping these issues in mind, we offer an empirical evidence on how planners could attract more travelers towards transit.

In this paper, we unravel the mode choice determinants while focusing on travel time, cost, and transit burden attributes in a transit-oriented mega city. In detail, we examine how travel time and cost difference between transit and car affects commuters' mode choice. Also, we develop models that can address the difference between socioeconomic groups regarding mode choice determinants. To do so, we employ the 2016 HTS data of the Seoul metropolitan area and other data sources including the web based route guide information. Based on our study, we elaborate policy interventions that can foster auto-to-transit mode shift while addressing the difference among the groups of commuters by income and age.

Findings suggest that travel time and cost difference between transit and auto clearly affects commuters' mode choice. Especially, the gap and ratio of travel time differences were significantly associated with mode choice regardless of the socioeconomic status of commuters. For example, results show that all else equal, a one minute increase in the gap of travel time can increase odds of driving by about 2.2–3.0%. Regarding the number of transfers, the results suggest that the odds of driving for commuters who have to transfer more than once when using transit are likely to increase by 24.3%. Interestingly, this odds increased more than 30% for the middle income and the youngest group. Finally, commuters turned out to consider walking during transfer as a great burden among other stages of walking. Regarding access and egress distances, the results show that travelers are more sensitive to egress distance.

2. Literature review

As a part of understanding travelers' behavior, a large portion of existing literature have focused on identifying the determinants of mode choice. Well-known determinants include personal and household attributes, while these factors are known to be further related with the travelers' attitude (Parkany et al., 2004; Kamruzzaman et al., 2015). Regarding individual characteristics, Wu and Hong (2017) showed that a traveler is likely to keep driving even when the transit infrastructure is improved, once he or she owns a car. Another study by Ko et al. (2019) showed that income levels are significantly associated with mode choice, further suggesting that policy interventions for modal shifts should target middle income groups. Also, O'Fallon et al. (2004) reported that commuters living with children are generally more constrained to drive. Built environments are also known to affect travelers' mode choice as the impacts of density, diversity, walkability, and design factors are well documented in numerous studies (Cervero, 2002; Buehler, 2011; Manaugh and El-Geneidy, 2011; Chakrabarti, 2017; Ding et al., 2017). As a part, a recent study by Guerra et al. (2018) showed that higher density, better transit supply, and less roadway are associated with less driving based on an analysis on Mexico urban areas.

On the other hand, trip characteristics, including travel time and cost as well as transit burdens are known to affect mode choice decisions (Asensio, 2002; Frank et al., 2008; Yang et al., 2018). Regarding the trip attributes, Stradling (2002) asserted that saving money, time, and effort are the underlying aspects that can help attract travelers to ride public transport. Additionally, he elaborated that burdens such as walking, waiting, and maintaining body posture during transit rides are related with the term "effort" (Stradling, 2002). Especially for commuting trips, savings related with time, cost, and transit burdens may be more important than shopping or

leisure trips regarding that commuting is a daily routine activity. That is, savings made by choosing a certain transportation mode would accumulate, further fixating commuters' mode choice.

Regarding the trip attributes, travel time of different modes were identified as significant predictors of mode choice (Cervero, 2002; Frank et al., 2008). Cervero (2002) showed that longer travel time when taking transit compared to driving reduces the odds of taking transit. Also, Collins and Chambers (2005) reported that the travelers' preference of transit significantly decreases as the travel time when taking transit exceeds more than 1.5 times of driving. In line with these studies, Frank et al. (2008) also indicated that transit riders are more sensitive to travel time than to transit fares, and that riders value waiting times more costly than in-vehicle time. Based on these findings, Frank et al. (2008) implied that transit ridership could be increased by enhancing the competency of transit travel time, while insisting that reduction in driving time may offset the impact of transit improvements. Recently, Chowdhury et al. (2015) explained the travelers' behavior to save travel time by showing that transit riders are willing to endure transfer if they could reduce both their travel time and cost.

Besides, a recent study by Chakrabarti (2017) implied that travelers are influenced by the ratio of transit and auto travel time, indicating that lower travel time savings via driving would induce people to ride transit. While several studies have addressed the travel time difference between transit and driving to explain mode choice results, there seem to be no certain method for measuring the difference. For instance, Cervero (2002) and Zahabi et al. (2012) adopted the absolute difference, while Chakrabarti (2017) used the relative difference. Meanwhile, Limtanakool et al. (2006) is worthy for mentioning in that they considered both the absolute and relative difference in travel time between rail and car. Their rationale of applying both indices is based on the premise that the impact of travel time ratio (i.e. relative difference) on mode choice would be different by the travel time gap (i.e. absolute difference) and vice versa. They also insisted that the difference in travel time between car and transit more affects commuting trips than other purposes such as business or leisure. On the other hand, Sweet and Chen (2011) showed that the travel time unreliability of driving modes may induce people to shift their mode to transit.

Next, travel cost difference between modes is also reported to affect mode choice decisions (Pauley et al., 2006; Redman et al., 2013; Pneumatikou et al., 2015). For instance, Redman et al. (2013) concluded that public transit could attract car users by modifying travel cost, implying that provision of promotional low-price transit fares could help abandon car using habits. Also, a recent study on commuters' mode choice in Atlanta showed that providing free transit could encourage transit use (Ghimire and Lancelin, 2019). In addition, Pauley et al. (2006) presented fare elasticities for different traveler types and showed that high income groups have the highest elasticity. They interpreted this result by explaining that high income groups have an alternative to choose, which is owning a car. Meanwhile, Asensio (2002) showed that altering travel cost is a weaker policy intervention tool compared to modifying travel times. Similarly, De witte et al. (2006) argued that free public transit does not affect travel behavior significantly. In spite of these findings, travel cost has not been always considered for modeling mode choice since the cost of transit modes usually do not vary within a limited study area (Chakrabarti, 2017).

Regarding travel time, one important aspect is the temporal variability in which travel time changes according to the time of departure. Most studies related to the temporal variability of travel time focused on transit modes, while emphasizing the variation in transit service and schedules over a day (Lei and Church, 2010; Lei et al., 2012). Moreover, the travel time for auto would also show temporal variability affected by the dynamics of road congestion levels. In this regard, it is plausible that the temporal variability in travel time would affect mode choice according to the time of departure even for the same origin and destination pairs. This concept of temporal variability could be also applied to travel cost regarding that there are early morning discounts and late-night bus with a surcharge in Seoul, Korea. This perspective of temporal variability in travel time was further emphasized in the concept of "travel time cube" (Farber and Fu, 2017).

To address the temporal variability in travel time, recent studies have applied web-based Application Programming Interfaces (APIs) as a way to estimate travel time on a real-time basis for various research objectives (Javanmardi et al., 2015; Kim and Chung, 2018; Rothfeld et al., 2019). For instance, Kim and Chung (2018) identified the travel time variability in Seoul by applying the Naver Map database. Also, Yang et al. (2018) applied the "route search" module provided by Baidu Map to examine whether travel times of chosen and non-chosen modes affect mode choice in China. More in detail, Naver Map and Baidu Map are on-line products, similar to Google Maps, while both rank the first in most-used search engine in South Korea and China respectively. Moreover, Chakrabarti (2017) used the Google Maps API to query the best transit route, and collected the data including transit travel time and the number of transfers required for every segment. Similarly, Ton et al. (2019) also employed the Google Directions API to measure travel times for non-chosen modes to examine whether travel time influences active mode choice. The application of the web-based route guide APIs is a recent trend in transportation analysis which enables to estimate the travel time for alternative modes (Javanmardi et al., 2015; Shakeel et al., 2016, 2017; Ma et al., 2017; Shabanpour et al., 2017; Golshani et al., 2018; Hasnine et al., 2019).

Another type of trip characteristic that affects mode choice is the transit burden, which includes access and egress walking, as well as transfers (Chapman et al., 2007; Shabanpour et al., 2017). Javanmardi et al. (2015) showed that longer access and egress walking distances when taking transit discourage people to use transit. Also, Gadepalli et al. (2018) showed that longer walking time for access and egress discourages travelers to use transit, while Yang et al. (2018) showed that travelers in the age between 30 and 50 years old are likely to avoid using transit when walking time during transfers are long. They explained this result by assuming that middle-aged travelers generally have higher time value and therefore prefer easier and time-saving travel modes. Another study by Shabanpour et al. (2017), which focuses on the nonmandatory trips (e.g. shopping, civic activities, etc.), showed that travelers more consider about the egress walking time when selecting modes. Additionally, Chakrabarti (2017) also showed that the number of transfers are less for the routes which travelers used transit compared to the routes which travelers decided to drive. Regarding the number of transfers, Liu et al. (1997) emphasized that the value of transfer penalty is larger than that of the long transit intervals. Also, Guo and Wilson (2004) reported that the average transfer penalty is equivalent to 9.5 min of walking based on their analysis on

the subway system in downtown Boston. On the other hand, a recent study by [Manaugh and El-Geneidy \(2015\)](#) showed that travelers tend to avoid transfers, while further elaborating that self-selected transit users are less sensitive to the number of transfers.

On the other hand, numerous studies have paid attention to the varying characteristics among different socioeconomic factors, such as age, gender, and income levels ([Cheng et al., 2017](#); [Figueroa et al., 2014](#); [Scheiner and Holz-Rau, 2012](#)). Especially, it is reported that each socioeconomic group shows different preferences, attitudes, habits, and even ecological norms, which all leads to diverse travel outcomes ([Scheiner and Holz-Rau, 2012](#); [Hjorthol, 2008](#)). For instance, [Cheng et al. \(2017\)](#) showed that policy interventions could focus on designing high-quality walking environment, which could better satisfy low-income commuters with high preferences on convenience. In addition, [Chang \(2013\)](#) focused on the mode choice for airport access by age groups and showed that elderly travelers concern more about “ticket price”, while non-elderly people consider more about “no transfer needed” and “punctuality”. These studies imply that it is important that policymakers understand the differences between diverse socioeconomic groups when formulating transportation policies. Meanwhile, recent two studies have focused on the income difference between travelers to better address equity issues ([Cui et al., 2019](#); [Allen and Farber, 2020](#)). While the both studies have employed the stratified model approach to examine the difference between low-income and high-income groups, [Cui et al. \(2019\)](#) suggested implications that could help low-income groups reduce their commute time. To add, [Allen and Farber \(2020\)](#) showed that improvements in accessibility could result in higher activity participation rates, especially for the low-income and car-less households. These studies show that understanding the main concern of low-income travelers, as well as devising transportation policies based on their demand could lead to better equity levels.

From previous studies, we could acknowledge that travel time, cost, and transit burdens are explicit determinants of mode choice decisions. Especially, the literature review suggests that travelers consider the competitiveness of each transportation mode and also tend to avoid burdensome activities such as walking and transfer activities when using transit. However, we still find some knowledge gaps that should be further addressed. First of all, there are not many studies that have dealt with the impact of travel time and cost difference between transit and auto on mode choice from both absolute and relative perspectives. Given the fact that it is now possible to consider more accurate travel characteristics using the web-based APIs, we can better investigate how travel time and cost differences could have an impact on mode choice. Although it is known that both travel time (cost) gap and ratio affects mode choice ([Limtanakool et al., 2006](#)), most recent studies have only focused on the ratio part ([Chakrabarti, 2017](#)). Second, we can better understand the influence of transit burdens on mode choice. While it is reported that longer walking distances and more transfer activities discourage transit use, there were not much attention to walking during transfer. By considering the stage of walking, we can gain further insights in order to devise policies to reduce walking activities for transit uses. In addition, most studies have set transfer activities as quantitative variables, even though travelers may not perceive it as a quantitative characteristic. Third, we also have knowledge gaps on how these travel time, cost, and transit burdens could affect commuters by different socioeconomic status, especially under a context of a transit oriented mega city. This issue is important regarding that planners from a transit-oriented city should further plan policies that could help maximize transit use effectively. Finally, given the enhancement in data sources (e.g. point-to-point travel survey data, route guide information on a real-time basis), we can acquire additional knowledge that cannot be obtained based on traditional data sources.

3. Study design

In this section, we explain the data sources applied in our analysis and the methodological approach. We then explain the details of the sample selection process and the specifications of the model. It should be noted that our analysis uses the binominal logistic regression model which allows us to build a standard discrete choice modelling framework. Finally, we present how the variables regarding travel time, cost, and transit burdens are estimated.

3.1. Data source

3.1.1. 2016 SMA Household Travel Survey (HTS)

The study area is SMA of South Korea, which is one of the transit-oriented cities in the east Asia (see [Fig. 1](#)). We used the disaggregated 2016 HTS data, which includes travel data of 309,398 people from 116,792 different households. In the survey, the heads of household are asked to answer their household characteristics such as income level and housing type and each individual is asked about the trips they have made during a weekday. In particular, trips made on Tuesday or Thursday are the target of the survey. The two weekdays were selected as they are considered typical weekdays. More in detail, excluding Monday and Friday is a conventional method because those two dates show a different travel behavior compared to others. Additionally, Wednesday is excluded for survey because it is designated as a “Family Day” every week in South Korea, which may further cause an effect on travel behavior. A third of the survey respondents reside in the Seoul city, while half and the remains live in Gyeonggi and Incheon area respectively. Particularly, the respondents were asked to fill out the specific address of their residence, as well as the destination of each trip. [Fig. 2](#) shows the spatial distribution of origin and destination points in the Gangnam area. As seen, the origin and destination locations of the respondents are spatially well distributed to analyze the influence of the access and egress distance on mode choice. These points were geocoded by using the SK Tmap “Full Text Geocoding” API and the address data. Additionally, only a limited number of studies were able to employ the point-to-point travel survey data with large samples ([Javanmardi et al., 2015](#)), while other studies have used data sets aggregated at a census-tract or traffic-analysis zone level ([Chakrabarti, 2017](#); [Ma et al., 2017](#)).

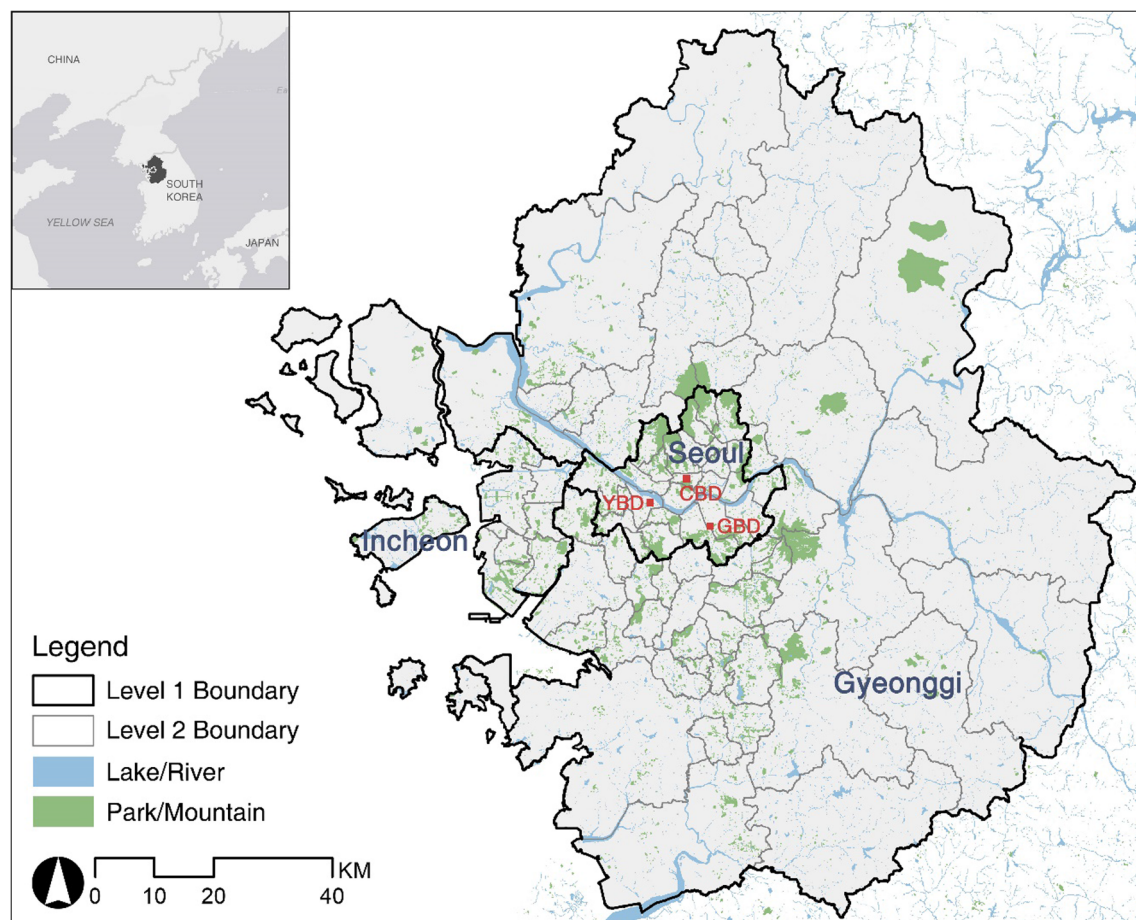


Fig. 1. Study area (Seoul metropolitan area). Note: Three city centers are presented (Central Business District (CBD); Gangnam Business District (GBD); Yeouido Business District (YBD)).

3.1.2. Web-based route guide information

To obtain information about travel time, cost, and transit burdens for each trip, we employed the web-based route guide information. The rationale for retrieving travel data from secondary data sources was to elicit travel information for both chosen and non-chosen mode. We used the SK Tmap route guide API to collect data for driving mode. SK Tmap provides an identical product with Google Maps, and ranks the first regarding the number of users in Korea. In addition, Google Maps does not provide any route search module for driving mode in Korea due to technical problem. This API service requires the location of origin and destination points, as well as the time of departure to ask travel data including time and cost (taxi fare) based on real-time information. The time of departure was obtained from the HTS in which respondents were asked to report.

For transit travel time and burdens, we employed the Google Maps Direction API which provides information about the fastest route for each origin-destination pair and departure time. By refining the data retrieved through Google Maps API, we measured the walking distances for access, egress, and transfer parts, as well as the number of transfers. Additionally, to collect data about transit cost, we applied the web-crawling technique on Naver Map. Naver Map is one of the most used websites in Korea, which is similar to Baidu and Google of China and U.S respectively. Our data collection process was done in March 2018, making time gap of 1.5 year with the HTS. In spite of the time gap, we judged that the results of this study are valid since transit infrastructure as well as major roads for driving in Seoul has not changed much during the period. Also, note that such time gap is inevitable for studies employing both web-based route guide information and HTS data (Javanmardi et al., 2015; Golshani et al., 2018; Chakrabarti, 2017; Hasnine et al., 2019).

3.2. Research approach

This study was designed to answer four research questions. In particular, the questions focus on the impact of travel time and cost difference between modes, as well as transit burdens on commuters' mode choice in Seoul. Details are as follows.

- First, we address whether both the absolute and relative differences in travel time and cost between modes impact commuters'

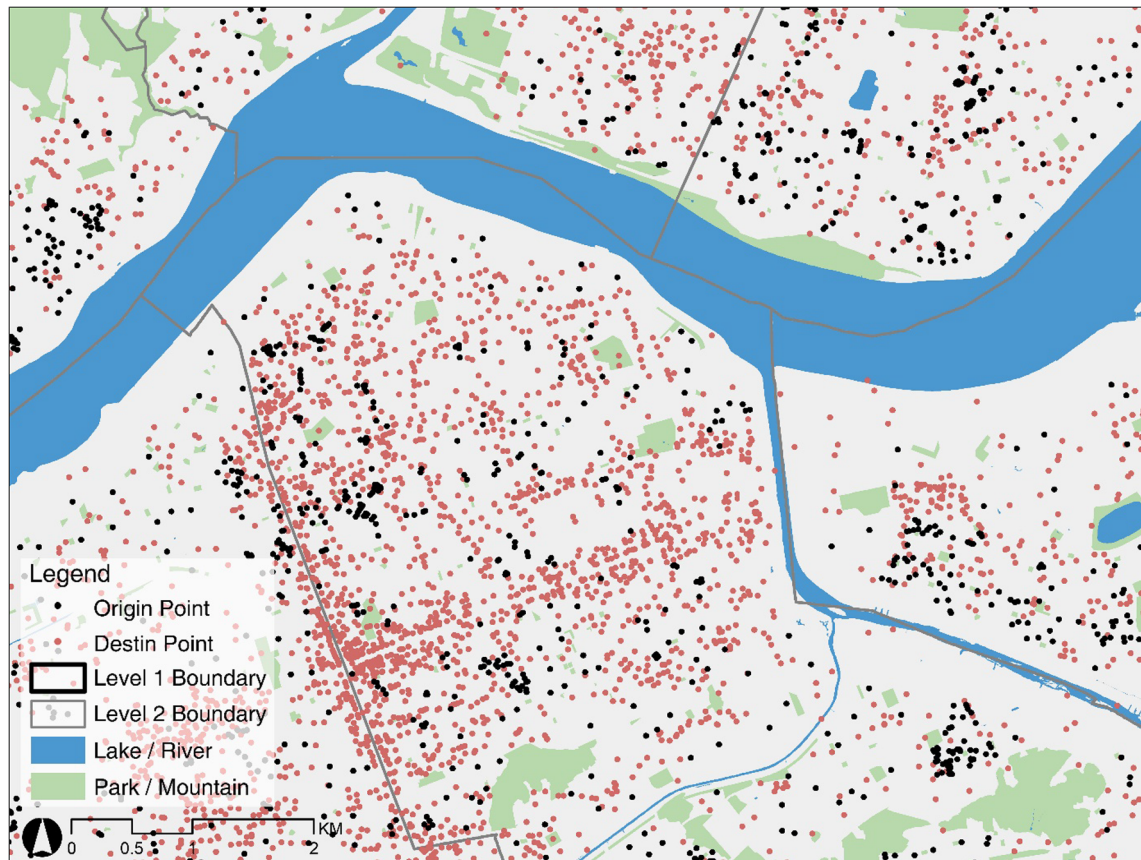
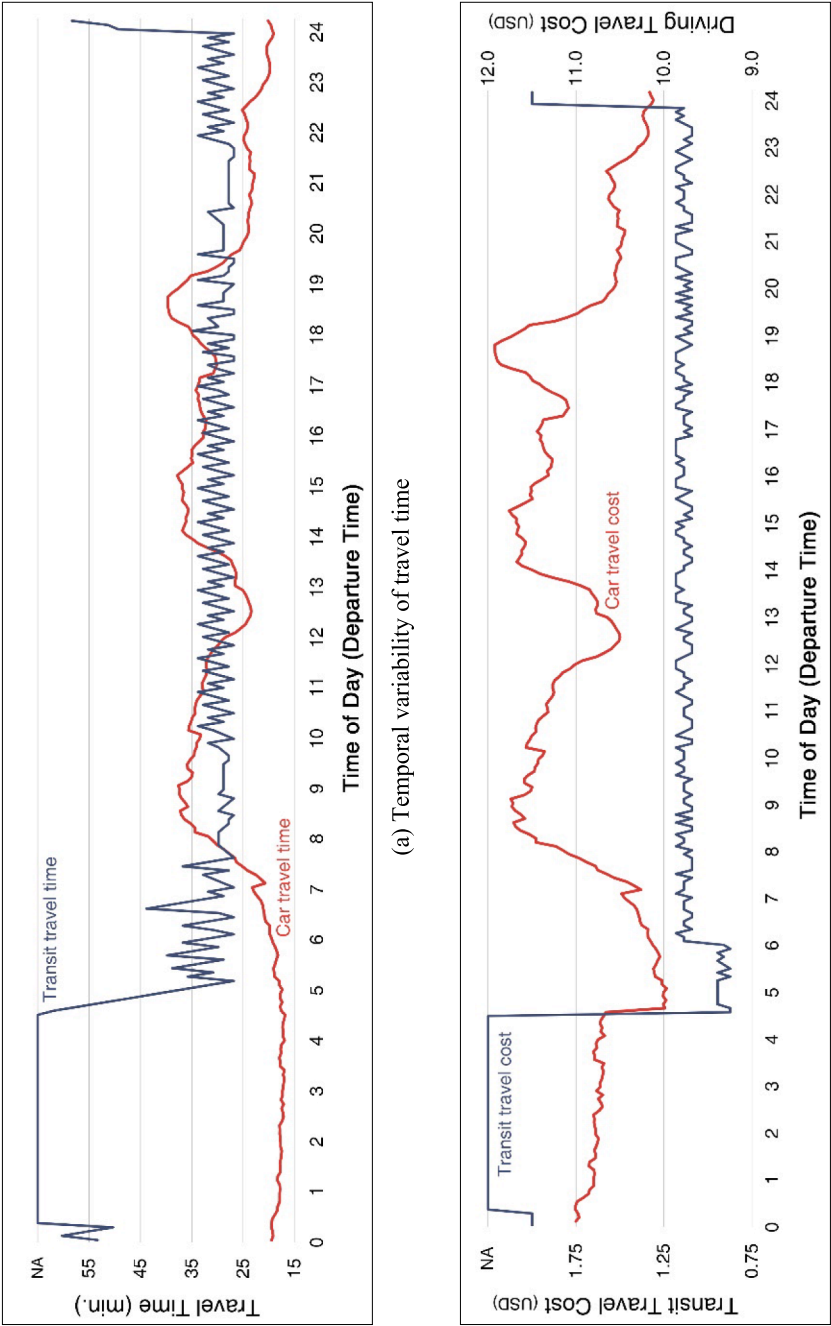


Fig. 2. Example of geocoded origin and destination points (Case: Gangnam area).

mode choice. Regarding this question, we apply an interaction term variable between the absolute and relative differences to examine whether commuters consider both aspects. We hypothesize that commuters select their mode based on the consumer choice theory, implying that mode choice results are done rationally by comparing the competency of two modes. In particular, [Limtanakool et al. \(2006\)](#) elaborated that both the gap and ratio should be accounted in the mode choice model. Additionally, we created an interaction term between gap and ratio to estimate the changing effect of ratio on mode choice as the gap changes. For instance, a unity increase in ratio would have a different impact on mode choice according to the amount of the gap.

- Second, we identify whether transit burdens, including access, egress, and transfer walking distances, as well as the number of transfers affect commuters' mode choice. Here, we hypothesize that commuters would less be affected by the access distance assuming that commuters are more familiar with the walking environment near their residence.
- Third, we hypothesize that commuters dislike longer transfer walking since it entails additional physical activity during their trip. For the number of transfers, we address whether commuters weigh the number of transfers orderly or nominally. We hypothesize that the number of transfers are perceived as nominal attributes by commuters and expect that commuters tend to drive when they have to transfer more than once.
- Finally, we focus on how the impact of travel time, cost, and transit burden on mode choice vary by different socioeconomic attributes such as age and income. Based on the results, we elaborate on illuminating policy interventions that can help commuters switch their modes to sustainable transportation modes.

Regarding our main variables, travel time and cost variables have their significances by addressing the temporal variability. In detail, the travel time difference between auto and transit varies according to the departure time, which may further affect commuter's mode choice. For certain routes, taking transit may have shorter travel time than driving during specific hours. This is well illustrated in [Fig. 3\(a\)](#) which shows an example of a particular route where using transit is faster than driving during morning peak hours. [Fig. 3\(a\)](#) is based on the web based route guide API, illustrating the fact that travel time for driving is affected by congestion levels, while travel time for transit is influenced by the transit schedule. In tandem with travel time, travel cost also shows temporal variability, while the variance is smaller. For transit travel cost, it varies according to the departure time since early morning riders benefit from promotions while they should pay more when riding nighttime bus (see [Fig. 3\(b\)](#)). These two figures show a conceptual picture on the changes in transit's competitiveness regarding time and cost values. In this regard, our analysis considers the temporal variability in the travel time and cost differences by using the departure time responded in the travel diary when querying the travel



(a) Temporal variability of travel time

(b) Temporal variability of travel cost

Fig. 3. Temporal variability of travel time and cost with departure time. Note: The origin and destination for this trip is designated as Wangsimni-station to Gangnam-station, which has a Euclidean distance of approximately 7 km. SK Telecom Route Guide API and Google Maps API are employed to retrieve data about travel time and cost.

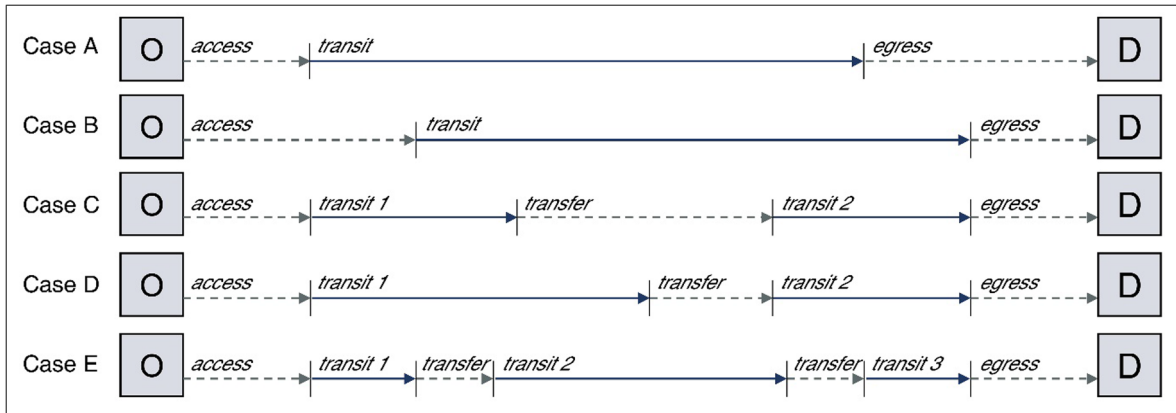


Fig. 4. Conceptual diagram of transit burdens.

characteristics through the web-based APIs.

For transit burdens, we address how walking distances and the number of transfers influence mode choice behaviors. As addressed in Fig. 4, walking distances by access, egress, and transfers part may vary by travel routes, even if the total walking distances are identical. For example, assuming that commuters avoid long walks during their egress, commuters might choose different modes for case A and B. Also, if commuters were to avoid long walks during their transfer, they might choose to ride transit for case D while prefer driving in case C. By examining how different parts of walking distance affects mode choice, we would possibly devise potential policy interventions in order to reduce walking distances which commuters react with high sensitivity. Regarding case E, we also focus on the number of transfers to further provide insights on how the connectivity of transit routes should be designed. In particular, we assume that the impact of walking burdens when using transit would differ according to the stage of walking. That is, when the walking is made is also important as well as the walking distance for the choice of travel mode.

3.3. Model specification

3.3.1. Sample selection

Using the 2016 SMA HTS data, we applied filters to decide which of the 600,126 trips can be utilized for this study. Trips that meet the following conditions were selected: (1) trips that are made by transit or driving, (2) individuals who hold a driver's license to assure that the individual had both transportation mode options, (3) households that own one or more vehicles, (4) home based trip to ensure that the commuter had to choose between transit and driving, (5) trip records that include the precise address for both the origin and destination points, (6) commuting trips, (7) individuals under the age of 65, (8) trips that were made when transit service is provided, and (9) trip records which the gap between reported travel time and the estimated travel time for chosen mode are less than 20 min. These filters were mainly employed to reassure that the individuals had chance to select their modes between transit and driving considering travel time, cost, and transit burdens. For example, individuals older than 65 were excluded because they benefit from the free public transportation policy. Also, for the last condition addressed above, we deemed that samples which show high discrepancy between reported and simulated travel time have low reliability. As a result, the final dataset applied in our study consists of 57,534 trips.

3.3.2. Discrete choice model

In this study, we employed the discrete choice model to examine how travel time, cost, and transit burdens influence commuters' mode choice. Regarding that we focus only on two transportation modes, driving and transit, we used the binomial logistic modeling approach. In particular, binominal logistic regression model is a standard form to analyze the effect of potential independent variables on a binary outcome variable (Chakrabarti, 2017). Also, logistic modeling approach is one of the most widely applied method in transportation studies, which has been numerous adopted (Frank et al., 2008; Ko et al., 2019; Ton et al., 2019). We have also considered applying the probit modeling approach, however, rejected based on the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) values which indicate the model performance. In the present study, the logistic regression model form for choice of driving (= 1) is given by:

$$\ln \frac{P(\text{driving} = 1)}{1 - P(\text{driving} = 1)} = \beta_0 + \beta_i X_i + \varepsilon$$

where β_0 is the coefficient of the constant term, β_i is the coefficient of the observed independent variable X_i , and ε is the error term. While minimizing AIC and BIC values are preferred, we also checked the multi-collinearity issue in our independent variables while excluding interaction terms.

3.3.3. Data segmentation

To achieve our research goal, we developed seven models. In our first model, we included all of the samples from the HTS. Then, three models were designed to analyze the mode choice determinants by income, while the other three model were devised to focus on the age difference. To do so, we created a subset of our data by household income level and age group, focusing on the distribution of samples. For income levels, we used the data from the HTS and divided 57,531 samples into three groups: low (24.6%), middle (43.6%), and high (31.8%) (see Table 1). In detail, the low income group's annual income is lower than 32,727 USD, while the middle group's income is between 32,727 and 54,545 USD and high income group earns more than 54,545 USD per year. For age groups, we segmented our data into three (see Table 1): under 35 (28.3%), between 35 and 50 (46.4%), and between 50 and 65 (25.3%). The rationale for this data segmentation process was to examine one of our research questions, which is to identify how transportation policies related to travel time, cost, and transit burdens may encourage transit ridership. It should be noted that we selected to use the stratified modelling approach instead of creating an interaction term for age and income variables to examine the age or income groups that show an insignificant relationship between mode choice and travel characteristics. This stratified modelling approach is widely adopted in recent transportation literature (Allen and Farber, 2020; Cui et al., 2019). In addition, we have two interaction variables for travel time and cost but interaction terms considering the three variables together were not considered due to the potential difficulty in interpreting the results.

3.3.4. Variable selection

The dependent variable of our study is the commuters' mode choice result. In detail, 0 and 1 refer to taking transit and driving respectively. The explanatory variables consist of individuals and household attributes, as well as our main variables which address travel time, cost, and transit burden characteristics. For individual attributes, we consider age, gender, job type, and whether each person is a household head. We also considered the square of age to identify how commuters select their travel mode as their age increases. For household attributes, we employed the housing type and income levels. Although income levels were surveyed as a categorical variable, we used it as a continuous variable. Meanwhile, these attributes were widely applied in previous studies, which are employed in our model as control variables.

For the variables related with travel time, we measured the absolute and relative difference in travel time between driving and transit. Here, we use both absolute and relative differences because using only one of the aspects may be inappropriate to precisely capture the difference. For example, assuming that the transit travel time is three times larger than driving, the absolute difference may be either 10 min or 40 min. The absolute difference is termed as travel time gap, while the relative difference is termed as travel time ratio. As aforementioned, we measured the travel time for both driving and transit by using web-based route guide information while controlling the origin-destination points and the time of departure for each trip (see Eqs. (1) and (2)). Additionally, we created an interaction term between the absolute and relative differences in travel time (Eq. (3)). The rationale for this interaction term is based on our hypothesis, while we assume that commuters value the travel time gap differently according to the travel time ratio. That is, a travel time gap of 20 min may impact commuters' mode choice differently when travel time ratio is 2.0 and 3.0.

In tandem with travel time, we also measured the absolute and relative differences in travel cost between two modes. Regarding that mode choices are a part of human decisions, it has been reported that people tend to consider both relative and absolute differences in price when choosing goods (Azar, 2011). We termed the two variables as travel cost gap and travel cost ratio, while the interaction term between the two measures was also employed (see Eqs. (4)–(6)). Travel cost by car was measured by the sum of taxi fare and the toll fee. For this, we assumed that taxi fare can reasonably capture all relevant costs of driving including fuel, maintenance, parking, tax, insurance and driving burden. Additionally, driving cost increases under congested conditions which makes it possible for the taxi fare to capture the congestion effect. It should be noted that using taxi fare to estimate driving cost does not mean that we view taxi identical to auto. Previously, Asensio (2002) used oil price to measure travel cost by car, however Limtanakool et al. (2006) asserted that it would be based solely on travel distance and would be highly correlated with travel time. It should be noted that travel time ratio and travel cost gap and ratio were transformed into natural logarithmic to improve the normality of the data. Meanwhile, it should be noted that including both ratio and gap variables did not raise an issue of any multicollinearity.

$$\text{Travel time gap} = \text{Travel time by transit} - \text{Travel time by car} \quad (1)$$

$$\text{Travel time ratio} = \ln (\text{Travel time by transit} / \text{Travel time by car}) \quad (2)$$

$$\text{Interaction term (Travel time)} = \text{Travel time gap} \times \text{Travel time ratio} \quad (3)$$

$$\text{Travel cost gap} = \ln (\text{Travel cost by car} - \text{Travel cost by transit}) \quad (4)$$

$$\text{Travel cost ratio} = \ln (\text{Travel cost by car} / \text{Travel cost by transit}) \quad (5)$$

$$\text{Interaction term (Travel cost)} = \text{Travel cost gap} \times \text{Travel cost ratio} \quad (6)$$

Regarding the variables for transit burden, we measured the walking distances for access, egress, and transfer part using the route guide information for each trip. For access walking, it refers to the walking distance from each residence to the transit station, while the egress walking refers to the distance from the transit station to each workplace. Transfer walking indicates the distance that should be walked during transfer activities. For the number of transfers for each trip, we applied it as nominal variables while 0 indicates no transfer, 1 indicates one transfer, and 2 refers to more than one transfers.

Table 1
Descriptive statistics.

Variable	Overall sample			Income group			Age group		
	(n = 57,531)	n/Mean	%/SD	Low income (n = 14,153)	Middle income (n = 25,057)	High income (n = 18,321)	Age < 35 (n = 16,268)	35 < = Age < 50 (n = 26,713)	50 < = Age < 65 (n = 14,550)
Mode choice									
Transit (= 0)	22,642	39.4%		6,048	9,567	7,027	8,928	9,350	4,364
Car (= 1)	34,889	60.6%		8,105	15,490	11,294	7,340	17,363	10,186
Age	42,596	10.25	57.3%	41,973	12,492	43,220	30,536	42,442	56,363
Gender	43,324	75.3%	10.75	10,745	19,495	13,084	10,590	20,750	11,984
Male [†]	14,207	24.7%	3.408	3,408	5,562	5,237	5,678	5,963	2,566
Female	32,511	56.5%	6.292	6,292	14,429	11,790	10,687	16,026	5,798
Job type	16,968	29.5%	4.941	4,941	7,245	4,782	4,252	7,383	5,333
White collar	7,315	12.7%	2.717	2,717	3,093	1,505	1,161	3,003	3,151
Pink collar (service, retail)	737	1.3%	203	203	290	244	168	301	268
Blue collar	39,281	68.3%	10.990	10,990	17,641	10,650	7,553	19,711	12,017
Other	0.221	0.52	0.220	0.220	0.55	0.179	0.311	0.271	0.029
Household head (others [†])	35,707	62.1%	7.007	7,007	15,571	13,129	9,215	17,461	9,031
Housing type	9,668	16.8%	3.116	3,116	4,132	2,420	3,006	4,058	2,604
Apartment [†]	11,271	19.6%	3.672	3,672	5,024	2,575	3,615	4,882	2,774
Single family	885	1.5%	358	358	330	197	432	312	141
Multi family	431	0.7%	431	431	—	—	128	154	149
Other	2,918	5.1%	2,918	2,918	—	—	927	1,049	942
Income level(Annual)	10,909–21,818 USD	18.657	16.20	19,229	18,520	18,403	17,207	18,945	16,301
0–10,909 USD	21,818–32,727 USD	10,804	18.8%	10,804	2,015	1,957	3,587	2,030	2,131
10,909–21,818 USD	32,727–54,545 USD	25,057	43.6%	—	25,057	—	6,691	12,556	5,810
21,818–32,727 USD	54,545–109,090 USD	16,275	28.3%	—	—	16,275	4,356	7,314	4,605
32,727–54,545 USD	109,090 – USD	2,046	3.6%	—	—	2,046	579	779	688
54,545–109,090 USD	Travel time gap (min.)	18,657	16.20	19,229	16,19	18,403	17,207	18,945	16,301
109,090 – USD	Travel time ratio	2,023	0.93	2,125	2,015	1,957	1,915	2,030	2,131
Travel time	Travel cost gap (USD) ^{††}	10,465	9.07	9,727	10,527	10,951	11,032	10,517	9,736
Travel cost	Travel cost ratio	7,899	4.41	7,576	7,943	8,087	8,250	7,916	7,474
Transfer	Number of transfer (= 0) [†]	33,979	59.1%	8,949	14,615	10,415	9,336	15,704	8,939
Transfer	Number of transfer (= 1)	20,009	34.8%	4,456	8,912	6,641	5,969	9,307	4,733
Transfer	Number of transfer (> 1)	3,543	6.2%	748	1,530	1,265	963	1,702	878
Walking distance	Access walking (km)	0.319	0.24	0.322	0.25	0.317	0.23	0.319	0.24
Walking distance	Transfer walking (km)	0.037	0.08	0.032	0.037	0.040	0.038	0.037	0.035
Walking distance	Transfer walking (km) ^{†††}	0.120	0.10	0.117	0.119	0.123	0.118	0.121	0.120
Walking distance	Egress walking (km)	0.304	0.26	0.312	0.301	0.302	0.298	0.307	0.306

[†] Variables designated as the criterion variable for categorical values.^{††} USD = 1100 KRW.^{†††} Average and standard deviation values for trips that require transfer.

4. Results

4.1. Descriptive analysis

This section addresses a brief overview of the attributes of variables for the overall sample as well as the sample groups by income and age. The results are presented in Table 1 and Fig. 5. For mode choice results, transit ridership was about 39.4% while driving accounted for 60.6% based on the overall sample. As seen in Fig. 5, transit mode share was the highest for the low income and younger age groups, while the ridership of middle and high income groups were identical. The older age group between 50 and 65 showed the highest gap between the ridership of two modes.

According to the overall sample, the average age was 42.6 and three quarters were male. More than half of the respondents replied that their occupation refers to white collar. Also, about two-thirds of the sample were household heads and the average number of children per household was 0.2. Regarding travel time related variables, the average commuter in SMA has to drive 18.2 min to work or ride transit for 36.9 min. For travel cost, the average gap between modes was about 10.5 USD, while travel cost by car was 7.9 times higher than transit on average. Among 57,531 commuting trip samples, 59.1% did not require any transfer when riding transit, while 34.8% of trip needed transfer and 6.2% required more than one transfer. For walking distances, the average commuter in SMA had to walk about 300 m during their first and last mile each. Also, when considering trips that required one or more transfers, the average walking distance during transfer was about 120 m.

Table 1 also shows the descriptive statistics by income and age groups. Although not verified by statistical analysis, some attributes showed an increasing or decreasing pattern. For instance, commuters from high income households tended to reside in apartments: 49.5% for low income group and 71.7% for high income group. Also, the travel time gap for low income group was 19.2 min, while middle and high income groups were 18.5 and 18.4 min. This indicates that transit services may be less favorable to low income commuters. However, it is also possible that low income commuters commute longer distances than other groups which may increase the travel time gap between transit and driving. These explanations can also apply to sample groups by age. On the other hand, assuming that all sample take transit, a higher percentage of middle and high income groups had to transfer more during their commute compared to the low income group. It can be explained by assuming that low income commuters have direct transit routes even though it takes longer.

It should be noted that the sample used for the regression analysis may not completely represent the commuters residing in SMA. For example, the modal split of transit and auto modes are 59.0% and 41.0% in our study area, while the samples used in this study show a modal split of 39.4% and 60.6% (see Table 1). Additionally, the number of male workers are approximately two times larger than female, while Table 1 indicates that the difference is three times, not two times in our samples. This situation may be attributed to the sample selection process that considers only individuals from households with cars and commuters who take transit or auto and with a driver's license.

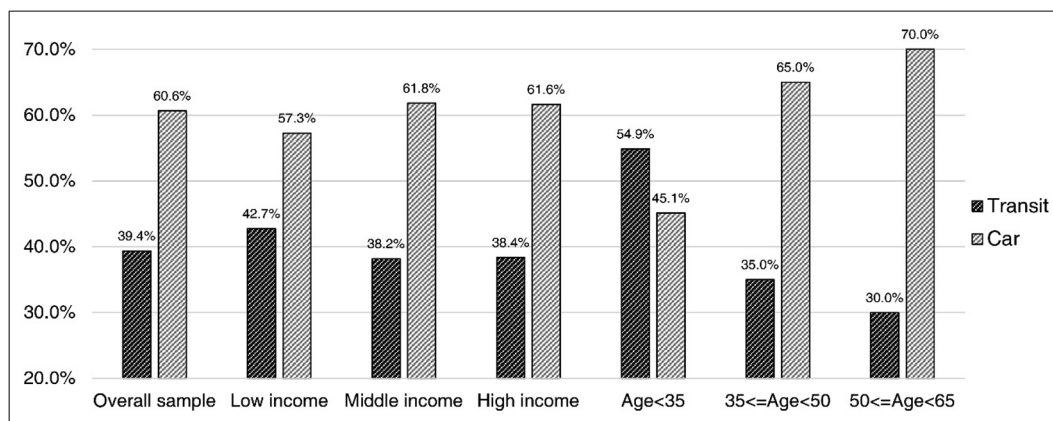


Fig. 5. Mode choice results by commuter groups.

4.2. Logistic regression analysis

4.2.1. Mode choice determinants

Table 2 presents the logistic regression model estimates for commuters' mode choice of the overall sample. The significance of the estimated model are supported by the Chi-square test. Among the individual and household attributes, age was strongly related with mode choice. As commuters get older, they tend to drive instead of riding transit, however, reverse their tendency at a certain point. Based on the coefficients, commuters showed the highest probability of driving when they are 53.4 years old. For gender, the results show that female commuters less tend to drive which is in line with previous studies (Limtanakool et al., 2006; Ko et al., 2019). The degree of car dependency also varied by the commuter's job type, while pink collar commuters showed the highest probability of driving. White collar workers, those who are likely to have their workplace in city centers where transit service is suitable, showed

Table 2

Logistic regression model based on the overall sample (0: transit, 1: car).

Explanatory variable		Overall		
		Odds Ratio	Coef.	z
Age		1.211	0.192	***
Age ²		0.998	−0.002	***
Gender (male = 0)		0.425	−0.856	***
Job type	Pink collar (service, retail)	1.038	0.037	*
(White collar = 0)	Blue collar	0.885	−0.122	***
	Other	2.681	0.986	***
Household head (= 1)		1.367	0.313	***
Number of children in household		1.178	0.163	***
Housing type	Single family	0.907	−0.098	***
(Apartment = 0)	Multi family	0.798	−0.226	***
	Other	1.051	0.050	
Income level	10,909–21,818 USD	1.187	0.172	
(0–10,909 USD = 0)	21,818–32,727 USD	1.348	0.299	***
	32,727–54,545 USD	1.561	0.445	***
	54,545–109,090 USD	1.681	0.519	***
	109,090 – USD	2.096	0.740	***
Travel time	Travel time gap	1.027	0.026	***
	Travel time ratio	1.666	0.510	***
	Travel time gap × Travel time ratio	0.992	−0.008	***
Travel cost	Travel cost gap	0.842	−0.172	**
	Travel cost ratio	0.483	−0.728	***
	Travel cost gap × Travel cost ratio	1.081	0.078	***
Transfer (= 0)	Number of transfer (= 1)	1.047	0.045	*
	Number of transfer (> 1)	1.242	0.217	***
Walking distance	Access walking dist.	1.082	0.078	*
	Transfer walking dist.	1.605	0.473	***
	Egress walking dist.	1.388	0.328	***
(Intercept)		0.006	−5.181	***
Model statistics	Number of obs.		57,531	
	LR chi2		9,258.63	***
	Log Likelihood		−33,934.56	
	AIC		67,925.11	
	BIC		68,175.99	
	Pseudo R2		0.120	

Note: Coefficients are based on the logit regression analysis.

* p < 0.10.

** p < 0.05.

*** p < 0.01.

less car dependency. Also, commuters who are household heads and have more children in their household showed higher car dependency. Especially, it is well known from previous studies that more children requires transportation mode that are faster (Kim and Ulfarsson, 2008; Ko et al., 2019). Regarding housing type, commuters who reside in apartment showed the highest possibility of riding car. This can be explained by the fact that apartment types are likely to have better parking conditions, which is in line with De Witte et al. (2013) and Ko et al. (2019).

Next, all of the variables related with travel time showed a strong relationship with commuters' mode choice. As expected, higher travel time gap and ratio, which implies longer travel time when taking transit, were associated with driving to work. For instance, a 10 min gap between transit and driving would increase the chance of driving a car by about 30.5%. We also found that the odds ratio of the interaction term between travel time gap and ratio was below 1.0, indicating that commuters are less affected by the travel time ratio as travel time gap increases. That is, when the travel time ratio increases from 2.0 to 4.0, commuters' tendency to drive significantly increases when the travel time gap is 10 min than when the travel time gap is 40 min (see Fig. 6(a)). In detail, the negative odds of the interaction term between travel time ratio and gap shows that the impact of travel time ratio on mode choice decreases as the travel time gap increases. Also, this result implies that commuters select their modes by valuing the travel time difference in both absolute and relative aspect. Based on this result, it is clear that improving transit's competency may enhance transit ridership.

For travel cost, the gap and ratio factors were negatively associated with driving. That is, commuters are likely to reduce their car dependency when the monetary cost for driving increases compared to transit. For instance, when the travel cost of driving is 7.39(=exp(2)) times higher than riding transit, the probability of driving is 48.2% of when the travel cost by car is 2.71(=exp(1)) times more expensive than transit. Also, the interaction term between travel cost gap and ratio exceeded 1.0 indicating that commuters' sensitivity to travel cost ratio decreases as travel cost gap increases. This interpretation is similar to the one done for travel time, which explains commuters' decision behavior regarding cost. The relationship among travel cost gap, ratio, and mode choice probabilities is addressed in Fig. 6(b). As seen in Fig. 6(b), the impact of travel cost ratio on mode choice is almost negligible when the

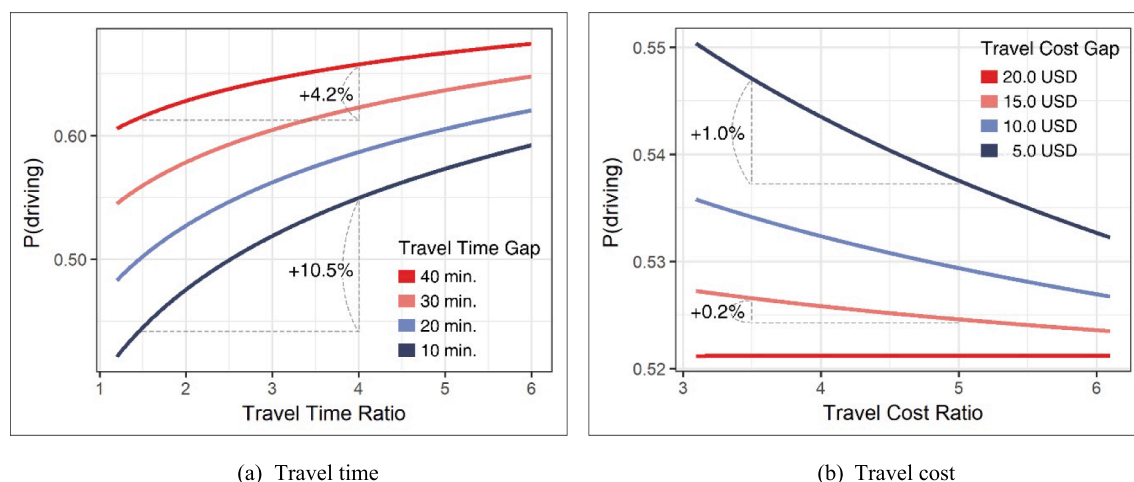


Fig. 6. Relationship between travel time, cost and $P(\text{auto})$.

travel cost gap is around 20.0 USD. Meanwhile, referring to the variance of the y-axis of both Fig. 6(a) and (b), commuters' possibilities of driving were more affected by travel time. This result is in line with Frank et al. (2008) and Zhou and Schweitzer (2011) which insisted that travelers are more sensitive to travel time than to transit fare costs when selecting modes.

As a part of transit burdens, the number of transfers required when taking transit showed a significant relationship with mode choice. Regarding that we employed the transfer variable as categorical values, the chance of driving to workplace increased by 4.7% when commuters needed to transfer once. Moreover, when commuters were required to transfer more than once, the car dependency increased by 24.2%. This result assures that commuters are not likely to ride transit when transfer is involved in their trip. Also, the odds ratio of the transfer variables imply that policymakers should focus on devising transit networks that enable travelers to accomplish their trip within one transfer. For example, more efforts could be done to reduce the number of transfers for trips that head to city centers or major employment areas from other parts of SMA. Although it is well known that travelers dislike transfer activities (Frank et al., 2008), this study contributes to the literature by addressing the fact that travelers do not perceive the number of transfers as continual terms.

Finally, the results show that commuters' mode choice is also influenced by the walking distance during access, transfer, and egress. Based on the odds ratio value, the probability of driving increases by $0.8 (= 8.2 \times 0.1)\%$ when the access walking distance increases by 100 m. Also, a 100 m increase in egress walking distance suggests a 3.3% higher chance of driving, while a 100 m increase during transfer increases the probability of driving by 4.9%. These results illuminate the fact that commuters are mostly influenced by the walking distance during transfer, while less affected by access walking. Besides, commuters were more sensitive to walking distances at their workplace than home when deciding their modes. Although Chakrabarti (2017) showed higher elasticity for transit stations near home than workplace, the results of our analysis are more identical with Grimsrud and El-Geneidy (2013) which reported that destination factors have greater effect on mode choice than origin. In addition, it is possible that the effect of walking distances on mode choice is smaller compared to previous studies. It is because SMA has a high density of transit stations making the effect smaller. In detail, the average of access and egress distances are 319 m and 304 m respectively, which are shorter than other regions such as the Greater Toronto and Hamilton area (865 m) or the Chicago metropolitan area (1260 m) (Hasnine et al., 2019; Shabanpour et al., 2017).

4.2.2. Mode choice determinants by income group

In this section, we mainly address the determinants of commuting mode choice by income group. Three models were examined, and the results are summarized in Table 3. Individual and household attributes showed a similar relationship with mode choice to the model of the overall sample. It should be noted that the variable for income level are omitted in these three models. For travel time variables, they were strongly associated with mode choice in all three models. However, the effect of the interaction term variable was not statistically significant for the high income group. One possible explanation for this result is that although high income commuters consider travel time when selecting modes, the robustness of the impact is lower than others regarding that they are likely to choose modes due to other reasons. For example, even if the travel time for driving does not guarantee sufficient time savings, high income commuters may prefer driving to enjoy comfortability or privacy.

For travel cost variables, low and middle income commuters turned out to consider the ratio and the interaction term, while the high income commuters were sensitive only to the travel cost gap. This result indicates that while high income commuters only consider the absolute cost they have to pay more, low and middle income commuters consider the relative difference much more. In other words, even if the travel cost gap is not high, low and middle income commuters more consider its relative amount compared to transit due to their limited budget. The odds ratio of the travel cost ratio for travel cost ratio was 0.330 and 0.387 respectively, indicating that low income commuters are more sensitive to travel cost. Regarding the travel cost factor, a recent study by Miller and Savage (2017) suggested that it is hard to make strong arguments about whether low income travelers are sensitive to it. They further

Table 3
Logistic regression model (Sample group by income level).

Explanatory variable		Low Income			Middle Income			High Income		
		Odds Ratio		z	Odds Ratio		z	Odds Ratio		z
Age		1.185	***	11.33	1.187	***	13.53	1.254	***	15.56
Age ²		0.998	***	−9.50	0.998	***	−11.39	0.998	***	−13.09
Gender (male = 0)		0.441	***	−17.85	0.448	***	−19.94	0.398	***	−20.40
Job type	Pink collar (service, retail)	0.948		−1.28	1.081	**	2.34	1.049		1.18
(White collar = 0)	Blue collar	0.805	***	−4.07	0.984		−0.34	0.837	***	−2.70
	Other	2.618	***	5.11	2.771	***	6.17	2.474	***	5.05
Household head (= 1)		1.118	**	2.39	1.531	***	11.14	1.346	***	6.48
Number of children in household		1.275	***	6.30	1.139	***	4.75	1.152	***	3.81
Housing type	Single family	0.768	***	−5.62	0.942		−1.52	1.037		0.70
(Apartment = 0)	Multi family	0.767	***	−5.98	0.805	***	−6.01	0.797	***	−4.66
	Other	0.842		−1.45	1.231	*	1.66	1.149		0.84
Income level										
Travel time	Travel time gap	1.027	***	7.52	1.026	***	9.71	1.026	***	7.96
	Travel time ratio	1.667	***	4.45	1.756	***	5.93	1.583	***	3.85
	Interaction term (travel time)	0.989	***	−4.41	0.991	***	−4.80	0.997		−1.13
Travel cost	Travel cost gap	0.867		−1.00	0.922		−0.74	0.754	**	−2.23
	Travel cost ratio	0.330	***	−2.97	0.387	***	−3.22	0.760		−0.77
	Interaction term (travel cost)	1.116	***	2.78	1.100	***	3.03	1.038		0.97
Transfer (= 0)	Number of transfer (= 1)	1.054		0.95	1.094	**	2.16	0.976		−0.50
	Number of transfer (> 1)	1.178		1.37	1.314	***	3.05	1.189	*	1.71
Walking distance	Access walking dist.	1.251	***	2.83	1.033		0.51	1.008		0.10
	Transfer walking dist.	2.080	**	2.06	1.204		0.72	2.113	**	2.51
	Egress walking dist.	1.444	***	4.89	1.434	***	5.77	1.265	***	3.09
(Intercept)		0.017	***	−10.70	0.012	***	−13.82	0.005	***	−14.58
Model statistics	Number of obs.	14,153			25,057			18,321		
	LR chi2	1841.96	***		3836.68	***		3657.21	***	
	Log Likelihood	−8739.12			−14743.14			−10369.06		
	AIC	17524.24			29532.28			20784.12		
	BIC	17698.07			29719.24			20963.88		
	Pseudo R2	0.095			0.115			0.150		

* $p < 0.10$.

** $p < 0.05$.

*** $p < 0.01$.

explained that low income travelers may be responsive to cost considering their limited budget, but also inflexible regarding their limited options.

Regarding the transfer variable, low income commuters were not influenced by the variable, while middle income commuters showed a strong association. On the other hand, high income commuters only tended to drive when they had to transfer more than once. In detail, this result suggests that the number of transfers were not a big problem for low income groups when selecting modes. One possible explanation is that low income commuters endure transfer burdens due to their limited budget. When it comes to the middle income group, transfer turned out to affect their mode choice. The middle income group is likely to have enough budgets, although limited compared to the high income group, to drive when they have to encounter higher burdens when taking transit. While the high income group is also likely to avoid transfers, they did not consider transferring once as a burden. These results can be also explained by the average Euclidean distance between the home and work by income groups. Low income group had the shortest distance of 8.1 km, while the middle and high income group had longer distance of 8.9 km and 9.4 km. This implies that low income group, traveling shorter distances, do not think transfer as burdens, while high income group consider transferring once as tolerable regarding their commuting distance. Most importantly, we can learn that middle income commuters' travel distances are a tipping point when transfer burdens can work as a great burden.

For walking distances, the probability of driving increased as egress walking distance increases regardless of income level. However, low income commuters were sensitive to both access and transfer walking distances, while high income commuters were influenced by the walking distance during transfer. Middle income commuters were not affected by both access and transfer walking distances. Regarding access walking distance, low income commuters may regard it significantly considering the environment around their residence. It is likely that walking conditions are poor for low income residential areas, further making low income commuters sensitive to access walking distance. This result is in line with Lee (2013) which showed that low income travelers consider the conditions of their environment as problematic. Also, for the walking distance during transfer, low income commuters were reluctant to longer walking distances although they were tolerable to the number of transfers. Transfer walking distance also affected high income commuters' mode choice, indicating that although they deem transferring once as admissible, they are still reluctant to walking. For the middle income group, the results suggest that their mode choices are solely determined by the number of transfers not walking distances during transfer.

Table 4
Logistic regression model (Sample group by age).

Explanatory variable		Age < 35		35 <= Age < 50		50 <= Age < 65	
		Odds Ratio	z	Odds Ratio	z	Odds Ratio	z
Age		1.365 ***	3.38	1.192 **	2.45	1.113	0.74
Age ²		0.996 ***	−2.58	0.998 **	−2.03	0.999	−0.86
Gender (male = 0)		0.443 ***	−20.18	0.441 ***	−21.12	0.333 ***	−18.66
Job type	Pink collar (service, retail)	0.989	−0.26	1.009	0.29	1.113 **	2.30
(White collar = 0)	Blue collar	1.232 ***	3.00	0.999	−0.03	0.673 ***	−7.45
	Other	2.588 ***	5.24	3.393 ***	7.01	2.011 ***	3.89
Household head (= 1)		1.482 ***	9.32	1.337 ***	7.90	1.107 *	1.72
Number of children in household		1.285 ***	7.91	1.094 ***	3.38	1.080	0.73
Housing type	Single family	0.868 ***	−3.00	0.920 **	−2.14	0.944	−1.11
(Apartment = 0)	Multi family	0.864 ***	−3.31	0.776 ***	−7.11	0.763 ***	−5.49
	Other	1.103	0.90	0.982	−0.14	1.031	0.16
Income level	10,909–21,818 USD	1.484 *	1.87	1.163	0.83	0.997	−0.02
(0–10,909 USD = 0)	21,818–32,727 USD	1.603 **	2.34	1.255	1.31	1.239	1.17
	32,727–54,545 USD	1.797 ***	2.92	1.525 **	2.46	1.383 *	1.81
	54,545–109,090 USD	1.748 ***	2.76	1.723 ***	3.15	1.529 **	2.34
	109,090 – USD	2.341 ***	3.88	2.067 ***	3.83	1.901 ***	3.19
Travel time	Travel time gap	1.022 ***	6.64	1.030 ***	11.58	1.027 ***	7.12
	Travel time ratio	2.056 ***	5.84	1.516 ***	4.62	1.606 ***	3.86
	Interaction term (travel time)	0.993 ***	−2.88	0.990 ***	−5.33	0.993 ***	−2.71
Travel cost	Travel cost gap	0.852	−1.24	0.833 *	−1.74	0.935	−0.44
	Travel cost ratio	0.675	−1.10	0.485 **	−2.53	0.366 **	−2.54
	Interaction term (travel cost)	1.059	1.52	1.083 ***	2.61	1.081 *	1.85
Transfer (= 0)	Number of transfer (= 1)	1.123 **	2.33	1.010	0.24	1.017	0.30
	Number of transfer (> 1)	1.334 ***	2.80	1.167 *	1.82	1.285 **	1.97
Walking distance	Access walking dist.	1.166 **	1.98	1.067	1.05	0.986	−0.16
	Transfer walking dist.	2.208 ***	2.63	1.354	1.22	1.408	0.94
	Egress walking dist.	1.547 ***	5.80	1.276 ***	4.15	1.398 ***	3.99
(Intercept)		0.001 ***	−5.26	0.010 ***	−2.96	0.139	−0.47
Model statistics	Number of obs.	16,268		26,713		14,550	
	LR chi2	2,855.32 ***		2,802.94 ***		1,448.68 ***	
	Log Likelihood	−9,770.83		−15,894.05		−8162.89	
	AIC	19,597.66		31844.10		16381.78	
	BIC	19,813.18		32073.50		16594.17	
	Pseudo R2	0.128		0.081		0.082	

* $p < 0.10$.

** $p < 0.05$.

*** $p < 0.01$.

4.2.3. Mode choice determinants by age group

Table 4 shows the estimates of logistic regression results by age group. The effect of individual and household attributes was mostly identical to the overall sample, while male commuters and household heads were strongly associated with high car dependency. Also, the estimate of income level indicated that high income commuters are likely to drive than low income commuters. For travel time related variables, all factors were strongly associated with commuting mode choice (see Table 4). This result suggests that the difference in travel time between modes is one of the significant factors when commuters decide their mode, regardless of age group. Thus, improving transit to reduce the travel time difference with driving would enhance the possibility of mode shift in commuting travel. All other things being equal, a 10 min increase in travel time gap increased the probability of driving by 24.3 (= 1.022¹⁰) % for commuters under the age of 35. On the other hand, commuters older than 35 and younger than 50, tended to drive 34.4% more when travel time gap increased by 10 min. This result is in line with Yang et al. (2018) which mentioned that middle aged travelers are likely to have higher values of travel time, therefore prefer time saving modes.

Meanwhile, the impact of travel cost factors on commuting mode choice varied by age group. As presented in Table 4, commuters younger than 35 were not sensitive to travel cost difference while commuters between the age of 35 and 50 were influenced by cost when selecting modes. Also, commuters above the age of 50 were only affected by travel cost ratio. This result suggests that car dependent commuters below the age of 35 are not likely to shift their modes even when cost-related interventions are applied. Furthermore, the result indicates that young transit commuters are somewhat constrained to ride transit due to their limited budget. On the other hand, the middle aged group was sensitive to travel cost implying that they are disposed to change their modes when monetary benefits are given. In fact, middle aged commuters, possessing higher values of travel time, may select faster modes when the travel cost difference between transit and car is not large. Commuters above the age of 50 were marginally affected by travel cost factors, while their estimates were similar to the low and middle income groups (see Tables 3 and 4). This result indicates that the response to cost attributes of commuters older than 50 are ambivalent. That is, they are likely to possess sufficient budget, but also more concerned about monetary cost savings at the same time.

Table 5
Summary of analysis results (Stratified model by income level).

Factors		Overall sample	Income level			Statistical significance of difference in odds ratio		
			(a) Low	(b) Middle	(c) High	(a) vs. (b)	(a) vs. (c)	(b) vs. (c)
Travel time	Gap	1.027	1.027	1.026	1.026			
	Ratio	1.666	1.667	1.756	1.583	*	**	**
	Interaction term	0.992	0.989	0.991	0.997		**	*
Travel cost	Gap	0.842	0.867	0.922	0.754	*	**	**
	Ratio	0.483	0.330	0.387	0.760		**	*
	Interaction term	1.081	1.116	1.100	1.038		**	*
Transfer (=0)	(=1)	1.047	1.054	1.094	0.976	*	*	*
	(> 1)	1.242	1.178	1.314	1.189	*	*	*
Walking distance	Access	1.082	1.251	1.033	1.008	*	***	
	Transfer	1.605	2.080	1.204	2.113	*	*	**
	Egress	1.388	1.444	1.434	1.265		**	*

Note: Odds ratio are bold-faced if it is statistically significant ($p < 0.1$).

* $p < 0.10$.

** $p < 0.05$.

*** $p < 0.01$.

For the variables related with transit burdens, commuters under the age of 35 showed a significantly different pattern from other age groups (see Table 4). All commuters, regardless of age group, had a higher chance of driving when they had to transfer more than once and walk longer distances for egress. However, commuters under the age of 35 were the only group that were sensitive to transferring once, and also to access and transfer walking distances. One possible explanation for this result is that younger commuters are reluctant to physical transit burdens, at least in the context of this study. Still, this result does not indicate that younger commuters ride transit less or older commuters are inclined to take transit. As shown in Fig. 5, younger commuters are more dependent on transit which is similar to the reports from other regions (Grimsrud and El-Geneidy, 2013; Melia et al., 2018; Delbosc et al., 2019). What this result implies is that younger commuters are more likely to ride transit when policies related to reducing transfer and walking distances are devised, while older commuters are not likely to be sensitive. In addition, this result is in line with Hasnine et al. (2019) which reported that younger individuals are more sensitive to access walking distances.

5. Discussion

From previous literature, we learned that encouraging people's modal shifts toward sustainable modes are desirable. Also, several studies have suggested that travel time, cost, and transit burdens are significant factors which transportation policies can focus on. In this regard, we attempted to unravel the impact of travel time, cost difference and transit burden on commuting mode choice. We also focused on revealing the socioeconomic group that are likely to change their modes when transportation policy interventions are made. This was to provide potential policies that can efficiently induce commuters' mode shift. It should be noted that our study particularly addresses the mode choice determinants that can encourage modal shifts, not increasing transit ridership itself. Tables 5 and 6 summarize our analysis results, describing the impacts of factors that can help commuters reduce their car dependency. On the

Table 6
Summary of analysis results (Stratified model by age).

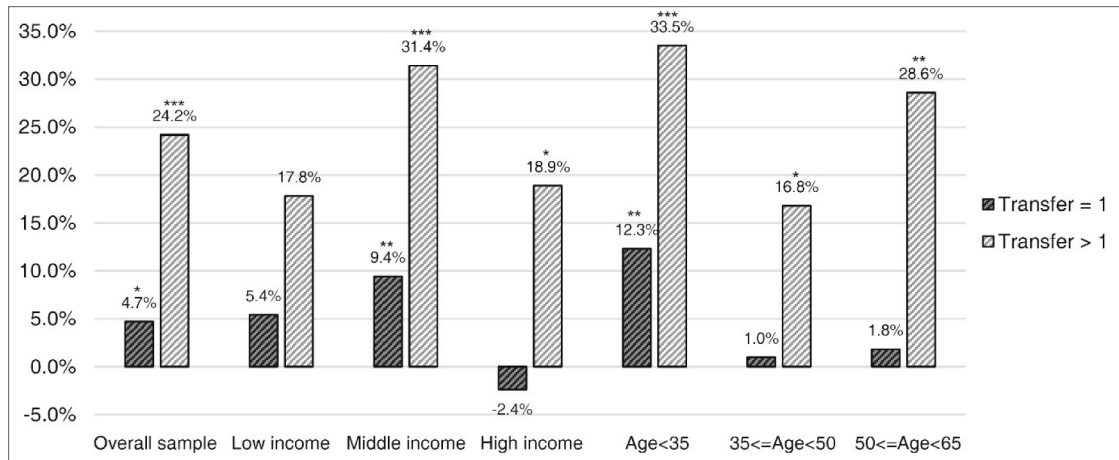
Factors		Overall sample	Age			Statistical significance of difference in odds ratio		
			(a) < 35	(b) 35 < = & < 50	(c) 50 < =	(a) vs. (b)	(a) vs. (c)	(b) vs. (c)
Travel time	Gap	1.027	1.022	1.030	1.027	*		
	Ratio	1.666	2.056	1.516	1.606	*	**	
	Interaction term	0.992	0.993	0.990	0.993		*	
Travel cost	Gap	0.842	0.852	0.833	0.935		**	*
	Ratio	0.483	0.675	0.485	0.366	*	***	*
	Interaction term	1.081	1.059	1.083	1.081	*	*	
Transfer (=0)	(=1)	1.047	1.123	1.010	1.017	**	**	
	(> 1)	1.242	1.334	1.167	1.285	**	**	*
Walking distance	Access	1.082	1.166	1.067	0.986	*	***	
	Transfer	1.605	2.208	1.354	1.408	**	***	
	Egress	1.388	1.547	1.276	1.398	*	**	*

Note: Odds ratio are bold-faced if it is statistically significant ($p < 0.1$).

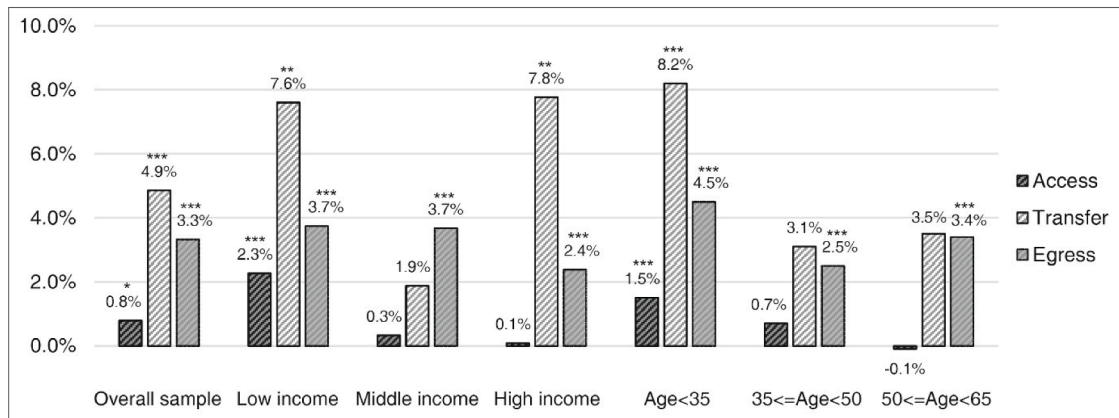
* $p < 0.10$.

** $p < 0.05$.

*** $p < 0.01$.



(a) Effects of number of transfers on mode choice (Reference variable = no transfer)



(b) Effects of walking distance on mode choice (100m increase in access, transfer, or egress)

Fig. 7. Effects of transit burden on mode choice by income and age group. Note: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

right side of each table, we noted the statistical significance of the difference in the odds ratio between the regression results from each subset sample. The statistical significance of the difference in the odds ratio was measured by using the interaction term variable (Buis, 2010).

As seen in Table 5 and 6, commuters were significantly affected by travel time difference between car and transit when selecting modes. This association was observed regardless of commuters' income level or age, indicating that improvement on transit service levels can help commuters get off their cars. For instance, assuming that the travel time ratio is 2.0, an increase in travel time gap by 10 min would increase the chance of driving by 29.5%. On the other hand, although travel cost difference was related with mode choice of the overall sample, the relationship turned out to be marginal when analyzed by income groups. This result is in line with Miller and Savage (2017) that explained the relationship between travel cost and transit ridership by income in two aspects. That is, lower income groups are likely to be sensitive to monetary cost values, while also being constrained to ride transit due to their limited budget. From this aspect, the analysis results show that younger commuters are more constrained, in other words compelled, to ride transit due to their limited budget, while commuters in their middle age are more sensitive to travel cost although their budgets are limited too. This implies that henceforth pricing policies can focus on promoting transit use of middle aged commuters.

For the number of transfers, most commuters were reluctant to transfer more than once, while low income commuters were not affected. That is, low income commuters are likely to suffer transit burdens because of their limited budget. Still, the results indicate that transportation policies should focus on enhancing the connectivity of transit infrastructures, so that commuters can travel within one transfer activity. For instance, when commuters have to transfer more than once when taking transit, the probability of driving increases by 31.4% and 18.9% for middle and high income commuters respectively (see Fig. 7(a)). Regarding age group of commuters, the youngest commuter group showed the highest reluctance to transfer. In particular, commuters under 35 were disinclined to endure transit burdens and showed that the probability of driving increases by 33.5% when they have to transfer more than once. This tendency was also observed for walking distances. While most commuters were unwilling to walk longer egress distance, the youngest commuter group was also sensitive to access and transfer walking distances. For instance, a 100 m increase in egress

walking distance would increase the probability of driving for commuters under 35 by 4.4% (see Fig. 7(b)). Also, when considering income, low and high income commuters were unwilling to walk longer distances during transfer. For example, high income commuters' probability of driving would increase by 7.8% if they had to walk 100 m more during transfer. In detail, low income commuters, who are likely to live in areas with poor walking environment, were sensitive to longer access walking. This result suggests policy interventions such as improving walking environment of low income residential areas, as well as reducing transfer walking distances.

6. Conclusion

This paper provides additional insights into the impact of travel time and cost difference between modes and transit burdens on commuters' mode choice. By applying discrete choice models, we also examined how commuters from different income and age groups react to travel time, cost, and transit burden attributes. First and foremost, we confirmed that commuters behave rationally when choosing transportation modes by taking account of both the travel time gap and ratio. In tandem with travel time factors, commuters were also responsive to travel cost difference, while some groups were constrained to ride transit due to limited monetary budget. Also, the number of transfers as well as the required walking distances for access, egress, and transfer showed a significant relationship with commuters' mode choice. To say the least, the results demonstrated potential levers for transportation policies to trigger modal shifts toward sustainable modes.

Considering the limited budget for policy interventions, we all acknowledge that we should devise policies that can generate the desired outcome effectively. Thus, there is no doubt that we should target a group of commuters when formulating policies, regarding their expected responsiveness. Beforehand, we suggest primary solutions for encouraging modal shifts, which may be effective regardless of income and age levels. All commuters were not disposed to transfer more than once and to walk longer egress distances. In this regard, transportation policies could focus their effort on enhancing transit connectivity so that commuters can access to major employment centers within one transfer. Also, policies could reduce the potential egress walking distance by increasing the density of transit stations in city centers. If applicable, improving the quality of transit so that the travel time difference between driving and taking transit could reduce is also a possible lever. Next, interventions on travel cost difference between modes may help encourage middle income commuters to ride transit. Based on our results, middle income commuters were those who are responsive to monetary cost factors. On the other hand, reducing access walking distance and further improving the walking environment of low income residential areas may also lead low income commuters to shift their modes. Additionally, younger commuters, although they ride transit the most, were reluctant to transit burdens, suggesting policies that can alleviate the burdens during transit ride. Based on these findings, transportation planners in Seoul, Korea may consider the main results of our study when providing the express train that aims to encourage transit use. In detail, they may design transfer systems that require short walking distances and consider the spatial distribution of commuters regarding their average age and income levels.

The contribution of this paper lies primarily in applying precise data for both chosen and non-chosen transportation modes to identify the impacts of time, cost, and transit burdens on mode choice, and second in its sample segmentation by income level and age. However, some limitations should be highlighted as well. First, our study is based on a cross-sectional data. Thus, the effect of policy interventions on modal shift cannot be addressed, suggesting further research efforts. Second, the data retrieved from the web based route guide information may be slightly different with reality. Still, this data approach is widely applied in recent studies, guaranteeing its use in transportation research (Chakrabarti, 2017; Yang et al., 2018; Kim and Chung, 2018; Ton et al., 2019). Regarding this issue, further study may also focus on analyzing the impact of perceived time, cost, and transit burdens on mode choice. Third, the driving cost measured in this paper may be underestimated due to other factors such as vehicle maintenance cost and auto burdens (e.g. walking to the parking lot, finding a parking space). Future works may focus on developing methodologies that can accurately estimate these factors, and further employ it in mode choice models. Fourth, we did not control the spatial autocorrelation issue that may take place in our variables. For instance, there may be a marginal level of spatial autocorrelation issue in housing types because apartment type residence is located in clusters. Finally, we were not able to address the service quality factors (e.g. convenience, crowding) of transit modes. This issue requires further analyses especially when using HTS with a large number of samples.

CRedit authorship contribution statement

Jaehyun Ha: Conceptualization, Methodology, Formal analysis, Writing - original draft. **Sugie Lee:** Supervision, Writing - review & editing. **Joonho Ko:** Resources, Writing - review & editing.

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